LIBSIGNAL: 交通信號控制深度強化學習的標準化框架

A Standardized DRL Framework for Traffic Signal Control

1 OVERALL DRL-TSC CONTROL PROCEDURE

1. Observe Traffic
SUMO/CityFlow lanes output queue length & density

→

2. Wrapper Convert
OpenAI Gym wrapper normalizes observation data

→

3. Agent Action
DRL Model selects optimal phase every 10s

→

4. Safety Guardrail
Inject minimum green duration & yellow transition

→

5. Simulate & Learn
Advance simulation, compute reward & train agent

Environment Observation (Step 1-2) | Agent Decision (Step 3) | Safety Constraints (Step 4) | Model Training (Step 5)

2 SYSTEM ARCHITECTURE LAYERS

Environment Layer

Supports mainstream traffic simulators (SUMO & CityFlow), decoupling simulation mechanics from DRL algorithms.

Wrapper Interface

Implements OpenAI Gym standard APIs. Standardizes data format of states, actions, and rewards.

Agent Layer

Hosts multiple standardized TSC reinforcement learning models (DQN, FRAP, CoLight, etc.) for fair comparison.

3 THE RL TRINITY (強化學習三要素)

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State Space (Observation)
• Lane Density (車道車輛密度)
• Current Phase (當前綠燈相位)
• Queue Length (各方向紅燈排隊長度)
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Action Space
• Decided every 10s (每10秒決策一次)
• Action: Select optimal phase from a set
• Forces 3-5s yellow light interval (強制安全黃燈)
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Reward Function

$$R = -\sum(\text{Queue Length} + \beta \times \text{Delay})$$

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// Guides agent to minimize queue length and delay
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4 DYNAMIC PHASE EXECUTION

Traditional Fixed-Time Control:
- Cycle: Phase A → Phase B → Phase C → Phase D
- Problem: High waiting times during off-peak hours

Dynamic Jumping (LibSignal)
Skip empty phases; jump directly to the direction with the highest traffic pressure.

Safety Guardrails
Enforce minimum green light duration (e.g., 10s) and yellow transition to prevent accidents.

5 STANDARDIZED TSC ALGORITHMS

DQN (Deep Q-Network)

Classic value-based model, maps observed traffic states to optimal phase actions.

FRAP

6 EVALUATION METRICS

Metric	Description	Impact
Travel Time	Average time to traverse network	Core KPI
Delay	Time lost due to stop & wait	Minimize
Throughput	Completed vehicles count	Maximize

7 ORCHESTRATION LOOP & PRACTICES

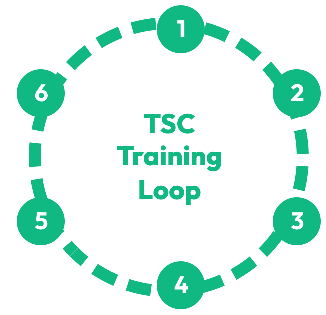
Models phase competition via relation networks, achieving invariant symmetry to intersection layouts.

CoLight

Employs graph attention networks (GAT) to coordinate control signals across multiple intersections.

Metric	Description	Impact
Queue Length	Max accumulation at red light	Minimize

- **Reproducibility:** Solves the 15% performance gap across simulators.



1. Obs State
2. Gym Wrap
3. Agent Select
4. Safety Guard
5. SUMO Step
6. Train Model